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Racial disparities in the energy burden beyond socio-economic inequality

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ABSTRACT

The consequences of climate change are known to be more damaging to historically minoritized groups. Ironically, recent measures to mitigate carbon emissions often negatively influence these groups to a greater extent. Racial and ethnic minorities are one of the well-documented energy vulnerable groups in the residential energy sector. Existing studies primarily attributed the racial and ethnic minorities' greater energy burden to socio-economic inequality, such as low income and inefficient housing, while other possible sources have received little empirical attention. Analyzing nationally representative U.S. households, we provide evidence that Blacks bear a greater energy burden than others unexplained by socio-economic inequality, and this additional energy burden is primarily driven by Blacks' extra demand for space heating. Specifically, after controlling for regional, housing, demographic, and appliance/device characteristics, Blacks' annual energy expenditure exceeds that of the other groups by US$1.6 billion. Examining this gap by end-use type and climate region, we reveal that this extra burden is primarily explained by Blacks' additional needs for space heating. Further evidence shows that self-reported needs for health services due to low indoor temperature are closely associated with the racial difference in space heating needs. Our results highlight a blind spot yet to be covered by existing energy policies and point to possible recognition injustice.

1. Introduction

Residential buildings are responsible for 10.9% of the global greenhouse gas emissions (Ritchie, 2020). Correspondingly, an ample body of research has examined the clean energy transition and demand reduction in the residential sector (Carley and Konisky, 2020; White and Sintov, 2020). Existing literature documents that climate change damages socially underrepresented groups to a greater extent (IPCC, 2014; Islam and Winkel, 2017). Furthermore, decarbonizing efforts to mitigate climate change often impose a disproportionate burden on these groups (Carley and Konisky, 2020; White and Sintov, 2020).

In this regard, to mitigate such inequalities, it is crucial to identify energy vulnerable groups and the source of vulnerability. In the energy context, vulnerable individuals are defined as those who "face a combination of non-negotiable energy needs as well as a lack of social and financial capital" (Middlemiss and Gillard, 2015). A growing body of research has contributed to identifying energy vulnerable groups, including low-income households, older people, disabled people, families with young children, and racial/ethnic minorities (Day and Hitchings, 2011; Anderson et al., 2012; Drehobl and Ross, 2016; Gillard et al., 2017; White and Sintov, 2020; Memmott et al., 2021). These energy vulnerable groups bear larger increases in energy bills under demand-response measures (White and Sintov, 2020), are more insecure under economic and energy crises such as the COVID-19 pandemic (Memmott et al., 2021), and have less access to advanced energy infrastructures (Brockway et al., 2021).

Among these vulnerable groups, our study focuses on racial and ethnic minorities. Existing studies have found that racial/ethnic minorities—especially, Blacks and Hispanics—bear a greater energy burden than other groups, and attributed this difference primarily to socio-economic inequality (Reames, 2016; Bednar et al., 2017; Hernández et al., 2016; Hernández and Siegel, 2019; Lewis et al., 2020). Specifically, due to unequal income and wealth distribution and residential segregation, racial/ethnic minorities tend to have houses and electronic devices with low energy efficiency, forcing them to spend more energy (Lewis et al., 2020; Friedman and Rosenbaum, 2004; Santamouris et al., 2007). Also, racial/ethnic minorities are more

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likely to be renters, and hence the split incentive problem discourages them from improving energy efficiency (Bird and Hernández, 2012).

Importantly, economic conditions and policies may distribute benefits and burdens in different ways across minority groups, resulting in further inequality among them. Recent literature has found that such racial inequality still remains significant even after controlling for demographics, housing conditions, and household income (Memmott et al., 2021; Kontokosta et al., 2020). However, little research has examined the sources of this remaining gap—especially, whether there exist any other sources of inequalities other than socio-economic inequality. In other words, the possibility that the equal distribution of socio-economic resources might not guarantee an equitable energy burden has been largely neglected.

Our paper provides unique evidence of this possibility: Black households bear a significantly greater energy burden than others with similar socio-economic backgrounds, and this extra burden is primarily driven by their greater needs for space heating. To demonstrate this, we analyze the Residential Energy Consumption Survey (RECS), which is widely adopted in the literature (Mohr, 2018; Karatasou and Santamouris, 2019). Our dataset with 5686 nationally representative U.S. households contains a rich set of socio-economic characteristics, such as regional, housing, demographic, and appliance/device characteristics, enabling us to control for these factors. Furthermore, it provides detailed estimates of households' energy usage and expenditure for each end-use type and their health needs related to indoor temperature. We leverage this advantage to decompose the racial difference in energy burden by the end-use type and examine related needs for health services, thereby identifying the sources of the gap.

We find three key results. First, we show a statistically and economically significant energy burden gap between Blacks and others inexplicable by socio-economic inequality. To be specific, after controlling for 56 variables related to socio-economic status, Black households' annual energy expenditure exceeds that of the other groups by US$120.2 on average. This implies that Black households in the U.S. annually bear an extra combined energy burden of US$1.64 billion. Moreover, the additional burden borne by Black households is larger in the high-income group than in the low-income group. This provides further evidence that the estimated gap is unlikely to be a mere consequence of socio-economic inequality, as the high-income group can afford investments in energy efficiency (Nair et al., 2010; Hernández and Phillips, 2015). Notably, Hispanic households do not present similar patterns, suggesting that this gap is likely associated with racial rather than ethnic differences.

Second, we demonstrate that Black households demand more space-heating energy than others, and this is the main driver of the racial gap in the energy burden. We find that whereas space heating accounts for only 25% of the total energy expenditure, it explains 52% of the racial gap in energy expenditure. By contrast, no other end-use type accounts for a greater share of the racial gap than its share of the energy expenditure. Notably, air conditioning explains only a negligible portion of the racial gap, although it accounts for about 10% of the total energy usage. In line with these results, the racial gap in space heating is greater in cold climates than in hot climates, whereas the gap in air conditioning does not exhibit any noticeable pattern.

Lastly, we provide suggestive evidence that Blacks' excess demand for space heating is associated with self-reported needs for health services due to low indoor temperature. Specifically, we find that the likelihood of self-reporting a need for medical attention due to low indoor temperature is higher for Blacks than the other groups with similar socio-economic backgrounds, while that due to high indoor temperature is statistically indistinguishable between Blacks and non-Blacks.

In sum, our results consistently show that the racial differences in space heating needs associated with contingent health impacts impose a significantly higher energy burden on Black households even when their socio-economic backgrounds are comparable to others. This suggests that Black households are doubly burdened: they not only face difficulties in affording a given level of energy needs due to socio-economic inequality, but also have higher energy needs under the same socio-economic conditions. The failure to address this inequity in policy design can lead to *recognition injustice*, defined as the lack of recognition of energy-vulnerable groups' differential needs (Jenkins et al., 2015; Sovacool and Dworkin, 2015; Walker and Day, 2012), which have received scant empirical attention in the literature (White and Sintov, 2020).

The remainder of this paper is organized as follows. In Section 2, we describe our dataset and variables. Section 3 estimates the racial gap in energy consumption and expenditure. Then, we disaggregate this racial gap to elaborate on the underlying mechanisms in Section 4. Section 5 examines the robustness of our results. Section 6 concludes the paper.

2. Data description

Our analysis relies on the 2015 RECS administered by EIA, designed to estimate energy consumption and expenditure for occupied housing units and households living in them. Each of the 5686 randomly sampled households was weighted to represent the U.S. population, including the residences not in the sample. In other words, the sampling weight of an observation is the number of households in the population it represents.

EIA constructed the dataset by combining the Household Survey with the Energy Supplier Survey. The Household Survey investigated energy-related characteristics and usage patterns of sample housing units. On the other hand, the Energy Supplier Survey collected data on the consumption of and expenditure on electricity, natural gas, propane/LPG, fuel oil, and kerosene in the sampled housing units during the reference year. To control the data quality, EIA compared household characteristics with billing and delivery data and revised them when obvious inconsistencies were detected.

The 2015 RECS identified consumption and expenditures for various end-use types (e.g., space heating, water heating, and air conditioning) in two steps. First, they estimated the expected consumption of each end-use type using engineering-based models, instead of statistical models used in the previous RECS. Specifically, engineering models improve upon statistical models building on the extant literature to construct physically principled models, which use published values for parameters such as estimates for Unit Energy Consumption. Second, they used the known total consumption to calibrate the engineering-model estimates by employing an optimization approach based on correlations and relative uncertainties of end-uses. To be specific, they used the Household Survey for appliance/device ownership, characteristics—e.g., size, age, Energy Star status, and usage frequency to estimate end-use amounts of consumption and expenditure. Then, they combined the initial estimates with the Energy Supplier Survey and calibrated them to reduce inconsistency. These steps were applied to each housing unit in the survey.

Dependent variables. The main outcome variables of interest are energy consumption (kBtu) and expenditure (US$) obtained from the Energy Supplier Survey. Given no zero values in these variables, we can log-transform them without additional treatment.

We consider three main end-use types—i.e., space heating, water heating, and air conditioning—and group the remaining end-use types as "the others". For space heating and water heating, we use total usage/cost across all fuel types for each end-use. Regarding air conditioning, we utilize electricity measures because RECS does not estimate air conditioning through other fuel types. We calculate the remaining usage and cost by subtracting the three end-uses from the total usage and expenditure. Since there are some zero values in end-use measures, we add one to these variables before the log-transformation. In Section 5, we check if potential distortions from this treatment lead to spurious findings and find consistent results.

Independent and control variables. Our independent variable is the householder's race, measured by the following question: "What is your race? Please select all that apply". The survey offered options adhering to the 1997 Office of Management and Budget (OMB) standards on race, such as "White", "Black or African American", "American Indian or Alaska Native", "Asian", "Native Hawaiian or Other Pacific Islander", and "Other (please specify)". 122 households selected multiple options; accordingly, RECS labeled these households as "Mixed".

We leverage a rich set of household responses collected from the Household Survey to capture potential confounding factors. First, we consider the household's ethnicity by including the variable "Hispanic", measured by the following question: "Are you Hispanic or Latino?" By doing so, we can separate the racial differences in outcome variables from the ethnic variation, which is considered a significant factor of energy poverty in previous studies (White and Sintov, 2020; Memmott et al., 2021; Reames, 2016; Wang et al., 2021).

Second, we capture several regional characteristics that could affect residential energy demand (Reames, 2016; Min et al., 2010). To address energy demand heterogeneity incurred by differences in temperature and humidity, we control for heating and cooling degree days in 2015 (which measure how cold or hot the temperature was), IECC Climate Code, census division, urban type, and the categorization of metropolitan/micropolitan statistical area.

Third, we include numerous house characteristics that substantially determine the household's energy efficiency (Reames, 2016; Bednar et al., 2017; Lewis et al., 2020; Lyubich, 2020): when a housing unit was built; when a household moved in; the numbers of bedrooms, full/half bathrooms, other rooms, and windows; types of glass in most windows, major outside wall material, and major roofing material; ownership of swimming pool, heated swimming pool, and on-site electricity generation from solar energy; the level of insulation; the frequency of draft; the total square footage; the type of housing unit; the number of stories; whether a housing unit is own or rent, studio or not, and has high ceilings or not. It is worth noting that housing characteristics alone explain over 50% (40%) of the variances in total energy consumption (expenditure).

Fourth, we control for household's demographic characteristics, which are associated with energy demand and unequal distribution of resources (Jenkins et al., 2015; Sovacool and Dworkin, 2015; Walker and Day, 2012): the number of household members; the number of household members with age 17 or younger; annual household income; a respondent's sex, age, education level, and employment status. In addition, we capture households' behavioral characteristics by controlling for the number of weekdays someone is at home, and who is responsible for paying for the electricity used.

Lastly, we further absorb the effects of appliance/device ownership and their Energy Star qualification, which reflect households' access to energy-efficient resources (Lyubich, 2020). Specifically, we capture the number of refrigerators, stoves, microwave ovens, televisions, desktop computers, laptop computers, tablet computers (or e-readers); having a dishwasher, clothes washer, clothes dryer; having an Energy Star qualified dishwasher, clothes washer, clothes dryer, refrigerator, freezer, lightbulbs, water heating, windows. One might raise a concern that selecting appliances might be endogenous. We directly address this concern by re-estimating the models by excluding these variables, and obtain highly consistent results (see Section 5).

We provide detailed descriptions and summary statistics for all variables in Tables A.1–A.3.

3. Estimating racial differences in energy demand

3.1. Econometric specification

To measure the racial disparities in energy consumption and expenditure, we estimate the following weighted regression models with a rich set of control variables:

$$Y_i = \beta_1 \cdot Black_i + \beta_{2\text{–}5} \cdot RaceDummies_i + \gamma \cdot Hispanic_i + \mathbf{Z}_i \cdot \Delta + \varepsilon_i \quad (1)$$

$$Y_i = \beta_1 \cdot Black_i + \gamma \cdot Hispanic_i + \mathbf{Z}_i \cdot \Delta + \varepsilon_i, \quad (2)$$

where Y_i is the dependent variable denoting either energy consumption or energy expenditure; $Black_i$ is a dummy variable indicating one if a householder is Black (or African–American), and zero otherwise; $RaceDummies_i$ is a set of dummy variables for the remaining races except White, the baseline group in Eq. (1); $Hispanic_i$ denotes one if a householder is Hispanic or Latino, and zero otherwise; $\mathbf{Z}_i$ is a set of control variables including regional, housing, demographic, and appliance/device characteristics; and ε_i is a heteroskedasticity-robust standard error.

In total, our model controls for 56 variables representing regional, housing, demographic, and appliance/device characteristics. Such an extensive set of control variables substantially absorbs the effects of energy demand determinants other than race—especially, the effects of socio-economic inequality. For instance, the extant literature suggested that climate, temperature, house area, insulation, housing tenure, household income, education level, employment status, family size, and appliance/device ownership can make disparities in energy usage and expenditure (Reames, 2016; Bednar et al., 2017; Lewis et al., 2020). Our model directly controls for these factors, enabling us to mitigate concerns about alternative explanations on our estimates.

Our main interest is β_1, the additional usage or expenditure of Black households compared to the baseline after accounting for the effects of potential confounders. Specifically, β_1 in Eq. (1) represents the difference between Black and White households, whereas β_1 in Eq. (2) represents the difference between Black households and all non-Black households. Also, γ represents the ethnic gap between Hispanic households and others after accounting for the householder's race and the controls.

3.2. Estimation results

Table 1 shows our estimates. We find that Black households' annual energy usage and expenditure are significantly higher than White households' by 6473 kBtu (8.2%) and US$109.7 (6.5%) in absolute (proportional) values, respectively. In contrast, other races' energy usage and expenditure are statistically indistinguishable from Whites', except that Asians' expenditure is lower than Whites'. Blacks' excess usage and expenditure remain largely the same when we change the baseline from White households to all non-Black households: Black households' annual energy consumption and expenditure are higher than the others' by 6729 kBtu (9.3%) and US$120.2 (7.1%). Because Black households in our sample represent about 13.6 million U.S. households, our estimates imply that at the national level, Black households bear an extra energy burden of US$1.6 billion per year compared to others with similar socio-economic backgrounds.

Notably, Hispanics, conventionally considered as ethnic minorities (Reames, 2016; Hernández and Siegel, 2019; Lewis et al., 2020), exhibit distinct patterns from Blacks. Their energy usage is statistically indistinguishable from the other groups, and their energy expenditure is significantly lower than the other groups. In other words, although both Blacks and Hispanics are disadvantaged by socio-economic inequality, only Black households remain to be overburdened after accounting for the effects of the control variables. These results provide additional evidence that the racial disparities in energy demand are unlikely to be attributable to the differences in wealth and resource accessibility.

Heterogeneous estimates across income groups in Table 2 further support our interpretation that socio-economic inequality is unlikely to be responsible for the estimated racial gap in energy burden. Specifically, if the estimated gap between Blacks and others were driven by unobserved confounders related to socio-economic inequality, the gap would be greater among low-income households than high-income households, because high-income households can better afford investments in energy efficiency and housing quality than low-income households (Nair et al., 2010; Hernández and Phillips, 2015). However, the subgroup estimates exhibit the opposite pattern: the racial gaps in energy usage and expenditure are greater among high-income households in terms of both absolute and proportional values.

Table 1
Racial differences in energy usage and expenditure.

Panel A: Using white households as the baseline (Eq. (1)).				
Dependent var.	Usage (kBtu/year)	ln(Usage)	Expenditure (US$/year)	ln(Expenditure)
Race:				
Black	6473***	0.0792***	109.7***	0.0628***
	(1584)	(0.0228)	(33.44)	(0.0195)
Asian	−3400	−0.0551	−187.6***	−0.0788***
	(2375)	(0.0382)	(48.69)	(0.0294)
Native	2191	−0.0167	−44.84	−0.0403
	(5650)	(0.0625)	(104.1)	(0.0491)
Pacific	−3954	−0.0949	157.4	0.118
	(5602)	(0.106)	(128.8)	(0.0901)
Mixed	−2357	−0.161	−0.389	−0.0334
	(2842)	(0.115)	(58.50)	(0.0656)
Ethnicity:				
Hispanic	−1177	−0.00923	−146.5***	−0.0750***
	(1450)	(0.0210)	(29.93)	(0.0183)
Regional char.	Yes	Yes	Yes	Yes
House char.	Yes	Yes	Yes	Yes
Socio-economic var.	Yes	Yes	Yes	Yes
Appliance/device var.	Yes	Yes	Yes	Yes
R-squared	0.625	0.635	0.589	0.616
Panel B: Using all non-Black households as the baseline (Eq. (2)).				
Dependent var.	Usage (kBtu/year)	ln(Usage)	Expenditure (US$/year)	ln(Expenditure)
Race:				
Black	6729***	0.0888***	120.2***	0.0684***
	(1580)	(0.0233)	(33.28)	(0.0196)
Ethnicity:				
Hispanic	−871.6	−0.00320	−129.4***	−0.0673***
	(1438)	(0.0210)	(29.80)	(0.0182)
Regional char.	Yes	Yes	Yes	Yes
House char.	Yes	Yes	Yes	Yes
Socio-economic var.	Yes	Yes	Yes	Yes
Appliance/device var.	Yes	Yes	Yes	Yes
R-squared	0.625	0.634	0.588	0.615

All models are adjusted by sampling weights. Robust standard errors are in parentheses.
Sample size: 5686. Number of Black households: 591.
*p < 0.1, **p < 0.05, ***p < 0.01.

Table 2
Heterogeneous racial differences by household income.

Panel A: Low-income households (annual income < US$60,000)				
Dependent var.	Usage (kBtu/year)	ln(Usage)	Expenditure (US$/year)	ln(Expenditure)
Race:				
Black	5957***	0.0722**	96.30**	0.0480*
	(1908)	(0.0294)	(37.92)	(0.0246)
Ethnicity:				
Hispanic	−1516	−0.00839	−116.8***	−0.0705***
	(1598)	(0.0273)	(35.08)	(0.0242)
R-squared	0.580	0.577	0.525	0.559
Panel B: High-income households (annual income ≥ US$60,000)				
Dependent var.	Usage (kBtu/year)	ln(Usage)	Expenditure (US$/year)	ln(Expenditure)
Race:				
Black	7938***	0.116***	185.1***	0.107***
	(2576)	(0.0327)	(65.62)	(0.0295)
Ethnic:				
Hispanic	−415.7	−0.000544	−142.5***	−0.0468*
	(2647)	(0.0317)	(52.54)	(0.0265)
R-squared	0.640	0.688	0.597	0.638

We use all non-Black households as the baseline, and include all controls in the main model.
All models are adjusted by sampling weights. Robust standard errors are in parentheses.
Sample size: 3147 (Panel A), 2539 (Panel B).
Number of Black households: 443 (Panel A), 148 (Panel B).
*p < 0.1, **p < 0.05, ***p < 0.01.

4. Decomposing racial gap in energy demand

4.1. Disaggregation of the energy demand gap by end-use type

To investigate the source of the unexplained racial gap in the energy burden estimated in Section 3, we disaggregate energy usage and expenditure by end-use type. Specifically, we consider three end-use types accounting for the largest shares of energy demand: space heating, water heating, and air conditioning. All remaining end-use types are grouped as "the others". Then, we re-estimate our model in Eq. (2) after replacing the dependent variable with energy usage and expenditure for each end-use type. In other words, we estimate the following weighted regression model:

$$Y_i^k = \beta_1^k \cdot Black_i + \gamma^k \cdot Hispanic_i + \mathbf{Z}_i \cdot \Delta^k + \varepsilon_i^k, \tag{3}$$

where Y_i^k denotes the energy consumption or expenditure for end-use type $k \in \{\text{space heating, water heating, air conditioning, the others}\}$. Our results in this section remain consistent when we estimate the disaggregated version of Eq. (1) that uses White households as the baseline group.

Table 3 presents the estimated coefficients of the Black indicator across different end-use types. For example, the coefficient in the first row and the second column is the estimate using the log-transformed space heating usage as the dependent variable. We find that Black households' excess energy burden is the largest for space heating both in terms of absolute and proportionate values. After controlling for socio-economic factors, Blacks' space heating usage and expenditure exceed the others' by 4067 kBtu and US$62.7 in absolute values, and by 35.2% and 24.4% in proportionate values. In contrast, Blacks' air conditioning usage and expenditure are statistically indistinguishable from or even less than those of the others. Black households also use more energy for water heating and the other end-use types, but these differences are markedly smaller than the corresponding gaps for space heating.

We emphasize that the large coefficient of space heating is not merely driven by its large share in the total energy demand. Fig. 1 clarifies this by comparing each end-use type's share in the racial gap in energy demand (i.e., β_1^k/β_1 in Eqs. (2)–(3)) with its share in the total energy demand (i.e., $\mathbb{E}[Y_i^k]/\mathbb{E}[Y_i]$).[2] It shows that space heating explains the largest and disproportionately large share of the energy demand gap. Specifically, whereas space heating accounts for only 43% (33,039 kBtu/77,723 kBtu) of the total energy usage, it is responsible for 60% (4067 kBtu/6729 kBtu) of the estimated racial gaps in the usage. Likewise, space heating explains only 27% (US$511.1/US$1882.3) of the total energy expenditure, while it accounts for 52% (US$62.7/US$120.2) of the expenditure gap.

In contrast, no other end-use type accounts for a disproportionately large share of the racial gap. Most notably, although air conditioning accounts for 8% and 13% of the total energy usage and expenditure, respectively, it makes only negligible contributions to the corresponding racial gaps. Also, the share of water heating in the racial gap is indistinguishable from its share in the total energy demand. Finally, the share of the other end-use types in the racial gap is smaller than the share in the total energy demand.

Our results show that Blacks have greater needs for space heating than others with similar socio-economic backgrounds, which is the main driver of the racial disparity in the energy burden. As mentioned in Section 3, our estimates are unlikely to have been driven by differences in energy efficiency due to socio-economic inequality, because our model controls for an extensive set of socio-economic characteristics. Furthermore, if the estimates were driven by unobserved confounders related to energy efficiency, energy usage/expenditure for air conditioning would also be higher for Blacks than for other races. However, air conditioning shows insignificant or even opposite patterns.

[2] Note that by linearity, $\sum_k \beta_1^k/\beta_1 = 1$.

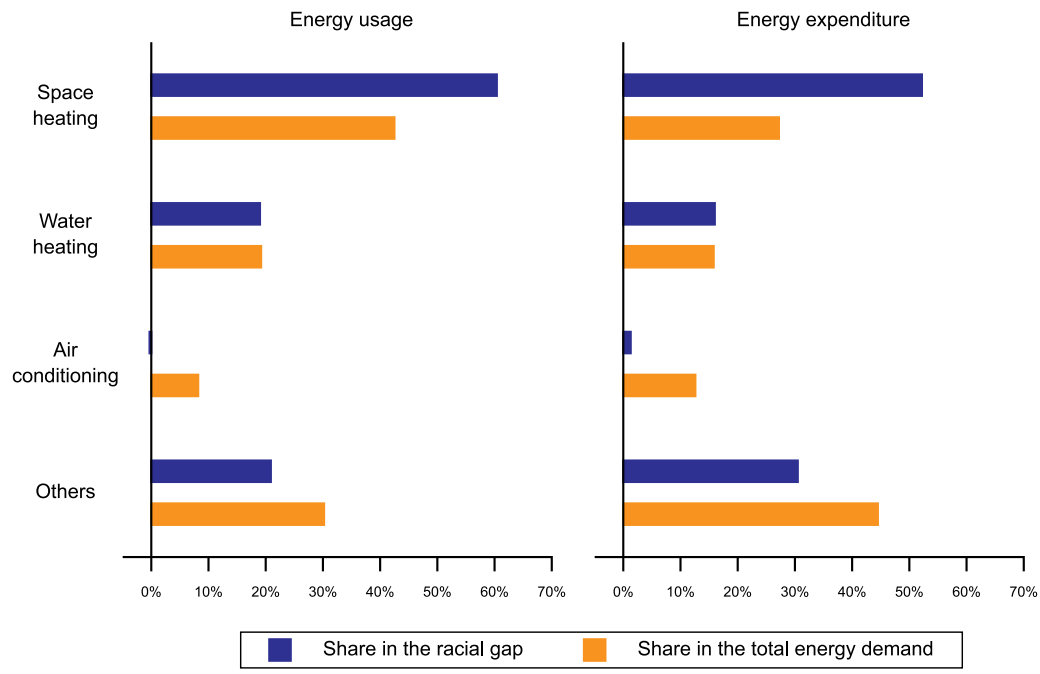

Fig. 1. Comparison between each end-use type's contributions to the energy demand gap and to the total energy demand. The black bars represent each end-use type's contribution to the energy usage/expenditure gap between Blacks and the others. The white bars represent each end-use type's share in the total energy usage/expenditure.

Table 3
Disaggregation of the racial gap by end-use type.

Coefficient: Black	Usage (kBtu/year)	ln(Usage+1)	Expenditure (US$/year)	ln(Expenditure+1)
End-use type:				
Space heating	4067***	0.302***	62.69***	0.218***
	(1167)	(0.0893)	(18.73)	(0.0583)
Water heating	1278***	0.0833***	19.27**	0.0522**
	(354.2)	(0.0293)	(8.896)	(0.0264)
Air conditioning	−21.92	−0.334**	1.58	−0.174*
	(277.3)	(0.150)	(9.625)	(0.0932)
Others	1406*	0.0299	36.63*	0.0209
	(725.9)	(0.0279)	(20.56)	(0.0256)

We use all non-Black households as the baseline, and include all controls in the main model.
All models are adjusted by sampling weights. Robust standard errors are in parentheses.
Sample size: 5686. Number of Black households: 591.
*p < 0.1, **p < 0.05, ***p < 0.01.

Table 4
Self-reported need for health services due to too low/high temperature.

Coefficient: Black	Medical attention needed because home was too cold	Medical attention needed because home was too hot
Sample:		
All	0.0139** (0.00685)	−0.00495 (0.00479)
Low-income	0.0206** (0.00905)	−0.00478 (0.00675)
High-income	0.00131 (0.00694)	−0.00593* (0.00326)

Each entry represents the Black indicator's coefficient in the corresponding regression. We use all non-Black households as the baseline, and include all controls in the main model.
All models are adjusted by sampling weights. Robust standard errors are in parentheses.
Sample size: 5686 (all), 3147 (low-income), and 2539 (high-income).
Number of Black households: 591 (all), 443 (low-income), and 148 (high-income).
*p < 0.1, **p < 0.05, ***p < 0.01.

4.2. Racial disparities by climate

Here, by exploring the interplay between end-use types and climate regions, we provide additional evidence that racial differences in space heating needs drive the energy demand gap. As the climate is the primary determinant of the demand for space heating and air conditioning (Reames, 2016; Min et al., 2010), we predict that the racial gap in space heating will be more prominent in cold climates than in hot climates.

Furthermore, we conduct a novel falsification test by examining whether the racial gap in air conditioning exhibits a similar pattern. If Black households' excess demand for space heating is indeed responsible for the energy usage gap, the racial gap in air conditioning will not exhibit any noticeable pattern. In contrast, if differences in energy efficiency due to socio-economic inequality are responsible for the energy demand gap, the racial gap in air conditioning will exhibit the opposite pattern to that in space heating. In other words, it will be more prominent in hot regions than in cold regions.

To distinguish between these competing explanations, we re-estimate Eqs. (2)–(3) after dividing the sample into three groups according to the climate: hot, moderate, and cold (see Methods). We categorize the climate regions according to the climate zones classified by the International Energy Conservation Code (IECC).[3] In our categorization, the hot region consists of Zones 1A, 2A, and 2B; the moderate region consists of Zones 3A, 3B, 4B, 3C, 4A, and 4C; and the cold region consists of Zones 5A, 5B, 5C, 6A, 6B, 7A, 7B, 7AK, and 8AK.

Fig. 2 summarizes our estimates. We find that the racial gaps in total energy usage and expenditure exhibit clear increasing patterns from hot to cold climates, and these patterns are mainly explained by space heating. Specifically, in hot regions, Black households' space heating usage and expenditure exceed those of the other groups by only 2040 kBtu and US$9. These differences are amplified as the climate becomes colder, and consequently become 7105 kBtu and US$100 in cold regions, statistically significant at the 0.05 and 0.10 levels, respectively. In contrast, the air conditioning demand presents economically and statistically insignificant differences between Blacks and others in all climates. These results are consistent with the explanation that the difference in space heating demand is the main driver of the racial gap in energy demand.

[3] Further details are available at https://codes.iccsafe.org/content/IECC2021P2/chapter-3-ce-general-requirements.

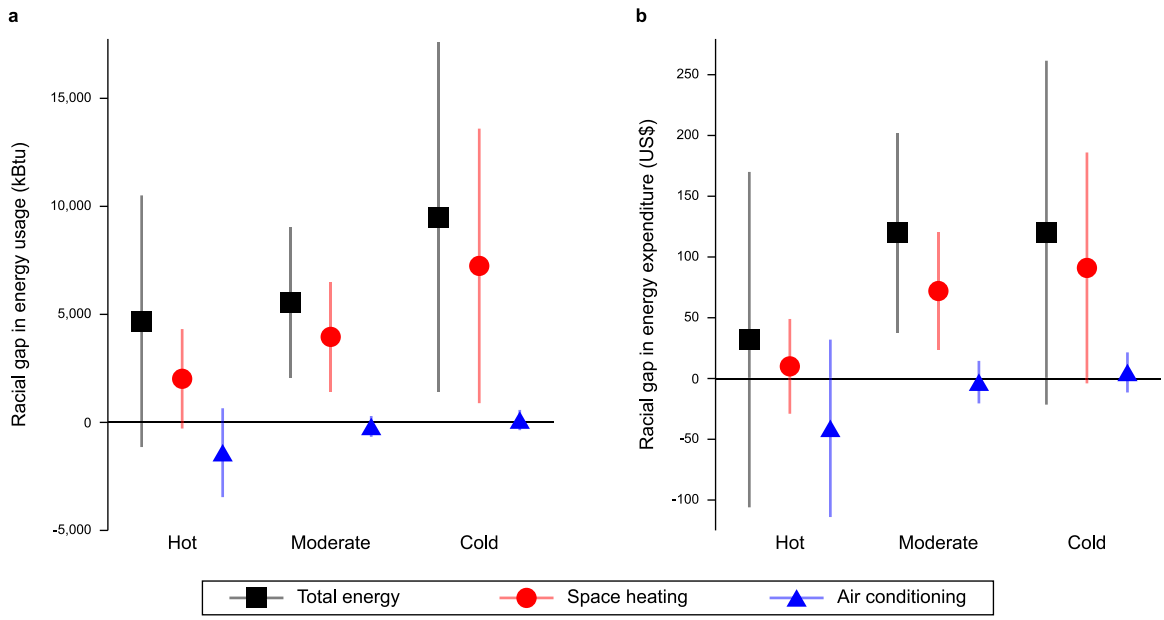

Fig. 2. Racial disparities in energy usage/expenditure by climate. The black squares denote the racial gaps in total energy usage/expenditure, the red circles denote those in space heating usage/expenditure, and the blue triangles denote those in air conditioning usage/expenditure. The vertical lines indicate 95% confidence intervals. Sample size: 952 (hot), 2726 (moderate), and 2008 (cold). Number of Black households: 110 (hot), 349 (moderate), and 132 (cold).

Table A.1
Summary of dependent and independent variables.

Variable	Description	Mean	SD	Min	Max
Dependent variables					
Usage	Total residential energy usage, in thousand Btu, 2015	77 722.89	46 962.31	201.574	490 187.4
Usage for space heating	Energy usage for space heating	33 039.20	33 492.61	0	277 614.80
Usage for water heating	Energy usage for water heating	14 891.03	9049.46	0	73 862.09
Usage for air conditioning	Energy usage for air conditioning	6338.70	7632.06	0	69 447.25
Usage for others	Energy usage for other end-uses	23 453.95	17 396.86	104.97	363 708.30
Expenditure	Total energy cost in 2015 (US$)	1882.34	1001.21	60.46	10 135.99
Expenditure for space heating	Energy cost for space heating	511.11	482.33	0	4766.66
Expenditure for water heating	Energy cost for water heating	296.66	208.84	0	2566.61
Expenditure for air conditioning	Energy cost for air conditioning	237.38	277.86	0	2860.01
Expenditure for others	Energy cost for other end-uses	837.19	564.67	31.48	7968.16
Medical attention (cold)	Medical attention needed because home was too cold	0.009	0.097	0	1
Medical attention (hot)	Medical attention needed because home was too hot	0.007	0.083	0	1
Independent variables					
Black	Householder (respondent) race is Black or African/American alone	0.104	0.305	0	1
Asian	Householder (respondent) race is Asian alone	0.043	0.202	0	1
Native	Householder (respondent) race is American Indian or Alaska Native alone	0.012	0.108	0	1
Pacific	Householder (respondent) race is Native Hawaiian or other Pacific Islander alone	0.004	0.062	0	1
Mixed	Householder (respondent) race is 2 or more races selected	0.021	0.145	0	1
Hispanic	Respondent is Hispanic or Latino	0.127	0.333	0	1

Table A.2
Summary of continuous control variables.

Variable	Description	Mean	SD	Min	Max
Regional characteristics					
cdd	Cooling degree days in 2015 (base temperature = 65 °F)	1719.2	1193.6	0	6607
hdd	Heating degree days in 2015 (base temperature = 65 °F)	3707.8	2149.3	0	9843
Housing characteristics					
bedrooms	No. of bedrooms	2.83	1.11	0	10
combath	No. of full bathrooms	1.75	0.75	0	6
halfbath	No. of half bathrooms	0.32	0.51	0	4
othrooms	No. of other rooms	3.36	1.62	1	14
windows	No. of windows	35.81	11.27	10	60
totsqrt	Total square footage	2081.4	1282.7	221	8501
swimpool	Have a swimming pool	0.08	0.27	0	1
pool	Have a heated swimming pool	0.03	0.16	0	1
solar	On-site electricity generation from solar	0.01	0.12	0	1
Household characteristics					
nhsldmem	No. of household members	2.57	1.43	1	12
nchild	No. of household members age 17 or younger	0.61	1.04	0	10
hhage	Respondent age	52.30	17.02	18	85
athome	No. of weekdays someone is at home	3.40	2.00	0	5
Appliance/Device characteristics					
desktop	No. of desktop computers	0.52	0.70	0	10
laptop	No. of laptop computers	1.04	1.06	0	10
tablet	No. of tablet computers or e-readers	0.98	1.12	0	7
frig	No. of refrigerators	1.40	0.68	0	8
freeze	No. of separate freezers	0.38	0.58	0	4
stove	No. of stoves	0.93	0.33	0	3
micro	No. of microwave ovens	1.01	0.30	0	7
dishwash	Have a dishwasher	0.71	0.45	0	1
cwasher	Have a clothes washer	0.85	0.35	0	1
dryer	Have a clothes dryer in home	0.84	0.37	0	1
tvcolor	No. of televisions	2.36	1.29	0	9
esdishw	Have an Energy Star qualified dishwasher	0.30	0.46	0	1
esfrig	Have an Energy Star qualified refrigerator	0.47	0.50	0	1
esfreeze	Have an Energy Star qualified freezer	0.12	0.32	0	1
esdryer	Have an Energy Star qualified dryer	0.36	0.48	0	1
escwash	Have an Energy Star qualified clothes washer	0.41	0.49	0	1
eslight	Have Energy Star qualified lightbulbs	0.40	0.49	0	1
eswater	Have an Energy Star qualified water heating	0.32	0.47	0	1
eswin	Have Energy Star qualified windows	0.25	0.43	0	1

4.3. Racial differences in the health impact of indoor temperature

Lastly, we provide suggestive evidence that Black households' additional needs for space heating might be associated with self-reported health conditions related to low indoor temperature. To this end, we regress two binary variables indicating whether a household needed medical attention in 2015 because of too low/high indoor temperature on the same set of independent variables as in previous sections. These two variables were measured using the following questions: "In the last year, did anyone in your household need medical attention because your home was too cold?" and "In the last year, did anyone in your household need medical attention because your home was too hot?" Then, respondents answered either "Yes" or "No" to each question.

Table 4 summarizes our estimates for self-reported health outcomes. After controlling for the effects of socio-economic factors, we find that Black households are 1.4%p more likely to report needs for health services due to low indoor temperature than others. In contrast, the need for medical attention due to high temperature is statistically indistinguishable between Blacks and the other groups, in accordance with our earlier result that the racial gap in air conditioning demand is statistically insignificant (Table 3). These findings suggest that the difference in health conditions associated with low indoor temperature, rather than overall health conditions or socio-economic statuses, is more likely to explain Blacks' excess demand for space heating.

In addition, we observe that the racial difference in the likelihood of medical attention due to low indoor temperature rises to 2.1%p in the low-income group (annual income < US$60,000), whereas it becomes statistically insignificant in the high-income group (annual income ≥ US$60,000). This is closely associated with our earlier result that the racial gap in energy demand is greater in the high-income group than in the low-income group (Table 2). Combining the earlier and current results suggests that high-income Black households can sufficiently afford to pay for their excess space heating demand, but low-income Black households are less likely to do so, which makes it difficult for them to avoid health concerns associated with low temperature. Consequently, while the inequality in the financial burden is more prominent in the high-income group, the inequality in the health outcome has a greater impact on the low-income group.

5. Robustness checks

We perform a series of robustness checks to test whether our findings are spuriously obtained by our specifications or sample selection.

Table A.3
Summary of categorical control variables.

Variables	Descriptions and Labels	
Regional characteristics		
iecc_climate	Description	IECC climate code
	Labels	1AL-2A (14.9%), 2B (1.9%), 3A (11.2%), 3B-4B (11.3%), 3C (3.7%), 4A (18.0%), 4C (3.8%), 5A (21.8%), 5B-5C (5.8%), 6A-6B (6.6%), 7A-7B-7AK-8AK (1.0%)
metromicro	Description	Housing unit in Census Metropolitan/Micropolitan Statistical Area
	Labels	Metropolitan (83.5%), Micropolitan (10.3%), Neither (6.3%)
division	Description	Census division
	Labels	NewL England (4.5%), Middle Atlantic (9.5%), East North Central (14.7%), West North Central (8.6%), South Atlantic (18.6%), East South Central (6.5%), West South Central (10.2%), Mountain North (4.0%), Mountain South (4.3%), Pacific (19.1%)
uatyp10	Description	Census 2010 Urban Type
	Labels	Urban Area (69.1%), Urban Cluster (10.5%), Rural (20.4%)
Housing characteristics		
yearmade	Description	Year when housing unit was built
	Labels	Before 1950 (15.1%), 1950–1959 (9.6%), 1960–1969 (9.9%), 1970–1979 (16.3%), 1980–1989 (15.4%), 1990–1999 (13.8%), 2000–2009 (15.6%), 2010–2015 (4.1%)
occupyyear	Description	Year when household moved in
	Labels	Before 1950 (0.3%), 1950–1959 (1.3%), 1960–1969 (2.1%), 1970–1979 (5.4%), 1980–1989 (7.6%), 1990–1999 (14.6%), 2000–2009 (27.8%), 2010–2015 (41.0%)
drafty	Description	Frequency of draft
	Labels	All the time (3.6%), Most of the time (6.9%), Some of the time (39.4%), Never (50.1%)
typeglass	Description	Type of glass in most windows
	Labels	Single-pane (38.6%), Double-pane (60.3%), Triple-pane (1.1%)
typehuq	Description	Type of housing unit
	Labels	Mobile home (5.0%), Single-family detached house (66.0%), Single-family attached house (8.4%), Apartment in a building with 2–4 units (5.5%), Apartment in a building with 5 or more units (15.1%)
adqinsul	Description	Level of insulation
	Labels	Well insulated (33.7%), Adequately insulated (48.9%), Poorly insulated (16.0%), Not insulated (1.4%)
ownrent	Description	Own or rent
	Labels	Owned or being bought by someone in your household (69.1%), Rented (29.8%), Occupied without payment of rent (1.1%)
stories	Description	Number of stories in a single-family home
	Labels	One story (43.6%), Two stories (27.4%) Three stories (1.5%), Four or more stories (0.04%), Split-level (2.0%), Not applicable (26.0%)
studio	Description	Studio apartment
	Labels	Yes (2.1%), No (18.5%), Not applicable (79.4%)
walltype	Description	Major outside wall material
	Labels	Brick (24.9%), Wood (16.3%), Sidling (34.2%), Stucco (15.0%), Shingle (composition) (2.4%), Stone (1.1%), Concrete or concrete block (5.5%), Other (0.7%)
rooftype	Description	Major roofing material
	Labels	Ceramic or clay tiles (4.1%), Wood shingles/shakes (4.6%), Metal (8.1%), Slate or synthetic shake (1.1%), Shingles (composition or asphalt) (64.0%), Concrete tiles (1.7%), Other (1.3%), Not applicable (15.1%)
highceil	Description	High ceilings
	Labels	Yes (34.7%), No (60.3%), Not applicable (5.0%)

(*continued on next page*)

Table A.3 (*continued*).

Household characteristics		
income	Description	Annual gross household income for the last year
	Labels	$0–$19,999 (17.4%), $20,000–$39,999 (22.2%), $40,000–$59,999 (15.8%), $60,000–$79,999 (13.4%), $80,000–$99,999 (8.7%), $100,000–$119,999 (7.44%) $120,000–$139,999 (4.87%), $140,000 or more (10.27%)
education	Description	Highest education completed by respondent
	Labels	Less than high school diploma or GED (7.1%), High school diploma or GED (23.6%), Some college or Associate's degree (33.3%), Bachelor's degree (20.8%), Master's, Professional, or Doctorate degree (15.1%)
employhh	Description	Respondent employment status
	Labels	Employed full-time (47.7%), Employed part-time (11.4%), Not employed/retired (41.0%)
elpay	Description	Who pays for electricity
	Labels	All used in this home paid by household (94.2%), All used in this home included in the rent/condo fee (3.4%), Some paid by household, some included in rent/condo fee (0.9%), Paid for some other way (0.6%)
hhsex	Description	Respondent sex
	Labels	Female (56.1%), Male (43.9%)

Table B.1
Estimation using different sets of control variables.

Dependent variable:	Usage (kBTU/year)			
Race:				
Black	−11,949***	−9158***	7029***	6857***
	(2312)	(2297)	(1604)	(1566)
Ethnic:				
Hispanic	−17,135***	−8611***	1241	−1680
	(1878)	(1892)	(1424)	(1443)
R-squared	0.019	0.151	0.587	0.609
Dependent variable:	ln(Usage)			
Race:				
Black	−0.185***	−0.143***	0.0884***	0.0877***
	(0.0340)	(0.0341)	(0.0253)	(0.0250)
Ethnic:				
Hispanic	−0.235***	−0.102***	0.0309	−0.0179
	(0.0288)	(0.0289)	(0.0221)	(0.0213)
R-squared	0.019	0.150	0.593	0.617
Dependent variable:	Expenditure (US$/year)			
Race:				
Black	−181.2***	−203.8***	115.5***	123.7***
	(45.34)	(45.75)	(35.18)	(33.73)
Ethnic:				
Hispanic	−407.0***	−288.0***	−87.25***	−154.8***
	(35.21)	(37.84)	(29.87)	(30.26)
R-squared	0.020	0.122	0.521	0.559
Dependent variable:	ln(Expenditure)			
Race:				
Black	−0.108***	−0.122***	0.0583***	0.0645***
	(0.0276)	(0.0276)	(0.0216)	(0.0207)
Ethnic:				
Hispanic	−0.237***	−0.151***	−0.0450**	−0.0849***
	(0.0240)	(0.0244)	(0.0191)	(0.0186)
R-squared	0.022	0.135	0.545	0.585
Regional char.	No	Yes	Yes	Yes
House char.	No	No	Yes	Yes
Socio-economic var.	No	No	No	Yes
Appliance/device var.	No	No	No	No

We use all non-Black households as the baseline.
Robust standard errors adjusted by sample weights are in parentheses.
Sample size: 5686. Number of Black households: 591.
*p < 0.1, **p < 0.05, ***p < 0.01.

First, we assess the sensitivity of our results by excluding and including different sets of control variables. We find that as long as house characteristics—a crucial determinant of energy efficiency—are included, the results remain quantitatively and qualitatively similar (Table B.1).

Second, there may be considerable variations in occupant behaviors across different racial groups. While these potential differences could result from racial disparities, we cannot rule out the possibility of reverse causality. For that reason, we analyze the racial gap in energy usage after accounting for these behavioral differences. As a proxy of how long residents stayed home, we include television consumption measures. Moreover, we control for the usage behaviors of space heating and air conditioning equipment. Table B.2 presents a summary of occupant behavior variables, and Table B.3 shows the estimated results after including these variables. We find that our main findings remain largely consistent, partially alleviating the concern about behavioral differences.

Third, our main specification does not consider the heterogeneity in the amount of heat generated from the same energy input (in Btu), depending on the fuel and the equipment used, which may bias our estimation. To address this issue, we re-estimate our model after taking account of the main fuel and equipment type for space heating. Table B.4 reports a summary of these variables, and Table B.5 presents the estimated results. We find that the coefficients remain qualitatively and quantitatively similar after capturing these confounders (Table B.5, row 1). In addition, our subgroup analysis by the main fuel type yields largely consistent results (Table B.5, rows 2–4).

Fourth, we address the possibility that the imbalance in covariates across races might lead to unreliable results due to model dependence by employing a propensity score matching procedure (Rosenbaum and Rubin, 1983; Datta et al., 2018). Specifically, we predict propensity scores of being Black by estimating a logistic regression model having $Black_i$ as a dependent variable and other covariates as independent variables. Then, we match each Black household to one non-Black household having a similar propensity score with a caliper size of 0.1. We find that this procedure substantially decreases the imbalance between Black and other households.[4] Using this sample, we re-estimate our models and find that all results are qualitatively unchanged (Tables B.6–B.8).

[4] The balance checks for the entire covariates are available upon request.

Table B.2
Summary of household behavior controls.

Variables	Descriptions and labels	
tvonwd1	Description	Most-used TV usage on weekdays
	Labels	Not applicable (2.4%), <1 h (3.3%), 1–3 h (26.1%), 4–6 h (36.7%), 7–10 h (18.2%), >10 h (13.4%)
tvonwe1	Description	Most-used TV usage on weekends
	Labels	Not applicable (2.4%), <1 h (2.7%), 1–3 h (17.8%), 4–6 h (33.3%), 7–10 h (25.5%), >10 h (18.3%)
tvonwd2	Description	Second most-used TV usage on weekdays
	Labels	Not applicable (27.1%), <1 h (14.0%), 1–3 h (36.3%), 4–6 h (14.4%), 7–10 h (5.2%), >10 h (3.1%)
tvonwe2	Description	Second most-used TV usage on weekends
	Labels	Not applicable (27.1%), <1 h (12.4%), 1–3 h (31.4%), 4–6 h (17.3%), 7–10 h (7.4%), >10 h (4.4%)
equipmuse	Description	Main heating equipment household behavior
	Labels	Set one temperature and leave it there mostly (37.9%), Manually adjust temperature at night or when no one is home (24.9%), Program the thermostat to automatically adjust temperature (17.1%), Turn equipment on or off as needed (13.4%), Does not have control over the equipment (2.0%), Other (0.2%), Not applicable (4.5%)
usecenac	Description	Central air conditioner household behavior
	Labels	Set ··· (29.2%), Manually ··· (17.3%), Program ··· (12.8%), Turn ··· (7.7%), Does not ··· (0.04%), Other (0.00%), Not applicable (33.0%) (Labels are the same as in *equipmuse*)
usewwac	Description	Most-used individual air conditioning unit household behavior
	Labels	Set ··· (7.8%), Manually ··· (4.6%), Program ··· (1.1%), Turn ··· (10.8%), Does not ··· (0.1%), Other (0.05%), Not applicable (75.5%) (Labels are the same as in *equipmuse*)

Table B.3
Estimation accounting for behavioral factors.

Coefficient: Black	Usage (kBtu/year)	ln(Usage+1)	Expenditure (US$/year)	ln(Expenditure+1)
End-use type:				
Total	6340***	0.077***	94.9***	0.053***
	(1552)	(0.023)	(32.4)	(0.019)
Space heating	4307***	0.233***	56.4***	0.177***
	(1158)	(0.055)	(18.7)	(0.043)

We use all non-Black households as the baseline.
We include all controls in the main model, and additionally control for variables in Table B.3.
All models are adjusted by sampling weights. Robust standard errors are in parentheses.
*p < 0.1, **p < 0.05, ***p < 0.01.

Fifth, we check whether the observed regional heterogeneity is restricted to the current categorization of climates. Recall that in Section 4.2, we categorize climate regions based on the IECC climate zones. To assess the sensitivity of our analysis, we alternatively adopt the Building America climate zone and re-estimate the heterogeneity across climates. We find that few households are categorized differently, and hence the estimates are quantitatively similar (Table B.9).

Sixth, we assess whether how we treat our dependent variables is responsible for the observed heterogeneity across end-use types. Specifically, we take the logarithm by adding one to the usage/expenditure variables before its transformation (the second and last columns in Table 3), which could distort the estimated elasticity. To address this concern, we employ the inverse hyperbolic sine transformation that allows us to retain zero-valued observations (Burbidge et al., 1988; MacKinnon and Magee, 1990; Bellemare and Wichman, 2020) and find qualitatively similar results (see Table B.10).

Finally, we examine if our findings on health outcomes are only observed in a particular model form. To be specific, in Table 4, we use a linear probability model with a heteroskedasticity-robust standard error, which yields unbiased and consistent estimates in a percentage point unit (Horrace and Oaxaca, 2006; Angrist and Pischke, 2008). We test the robustness of our findings by adopting alternative functional forms, such as logit and probit functions. Across different regression models, we obtain qualitatively consistent results (see Table B.11).

6. Conclusion

Taken together, we find a significant gap in the energy burden between Blacks and the other groups unexplained by socio-economic factors. Furthermore, this gap is mainly explained by Blacks' additional needs for space heating associated with contingent health impacts. This implies that Black households are doubly burdened: they not only face difficulties due to socio-economic inequality (Lewis et al., 2020), but they also present higher energy needs under the same socio-economic conditions. Thus, ensuring equality in socio-economic status does not necessarily ensure equity in the energy burden. The failure to account for such differential needs might impose a systematically higher burden on Black households, leading to recognition injustice.

Our results imply that there could be a blind spot not covered by existing policies on energy poverty. For instance, the U.S. federal government has implemented the Low-Income Home Energy Assistance Program (LIHEAP) and the Weatherization Assistance Program (WAP), aiming to improve the affordability of energy services and the efficiency of dwelling units, respectively (Carley et al., 2018; Bednar and Reames, 2020; Carley and Konisky, 2020). Prior studies have shown that these policies significantly improved recipients' health and energy security (Murray and Mills, 2014; Tonn et al., 2018). Notwithstanding the effectiveness of these policies on their targets, the inequity between Blacks and other races remains unresolved for the following reasons. First, even after achieving the same energy efficiency, Black recipients still have greater needs for space heating than other recipients. Second, high-income Black households, whom policies targeting low-income residents have neglected, bear a higher financial burden for energy consumption than other high-income households.

Table B.4
Space heating fuel and equipment types.

Variables	Descriptions and labels	
fuelheat	Description	Main space heating fuel
	Labels	Not applicable (4.5%), Natural gas (49.1%), Propane (4.2%), Fuel oil/kerosene (4.3%), Electricity (34.7%), Wood (cordwood or pellets) (3.0%), Some other fuel (0.1%)
equipm	Description	Main space heating equipment type
	Labels	Not applicable (4.5%), Central furnace (60.1%), Heat pump (12.3%), Steam/hot water system with radiators or pipes (6.0%), Built-in electric units (7.3%), Portable electric heaters (2.4%), Built-in room heater burning gas, oil, or kerosene (2.7%), Wood-burning stove (cordwood or pellets) (2.4%), Fireplace (0.7%), Built-in floor/wall pipeless furnace (1.0%), Some other equipment (0.6%)

Table B.5
Estimation accounting for the fuel type and heating equipment type.

Coefficient: Black Dep. var: Space heating	Usage (kBtu/year)	ln(Usage+1)	Expenditure (US$/year)	ln(Expenditure+1)
All sample	3335***	0.140***	55.8***	0.125***
	(1122)	(0.042)	(17.4)	(0.037)
Subgroups:				
Natural gas	5788***	0.133***	40.5*	0.09**
	(1952)	(0.044)	(20.8)	(0.043)
Electricity	284	0.120*	28.7	0.117*
	(630)	(0.063)	(23.0)	(0.060)
The others	5687	0.282	159.6*	0.233
	(4302)	(0.268)	(87.3)	(0.173)

We use all non-Black households as the baseline, and include all controls in the main model.
In row 1, we include the main fuel type and the main heating equipment type as controls.
In rows 2–4, we include the main heating equipment type as a control and restrict the sample according to the main fuel type.
All models are adjusted by sampling weights. Robust standard errors are in parentheses.
*p < 0.1, **p < 0.05, ***p < 0.01.

Table B.6
Racial differences in energy demand among the matched sample based on propensity scores (N = 976).

Dependent variables:	Usage (kBTU/year)	ln(Usage)	Expenditure (US$/year)	ln(Expenditure)
Race:				
Black	6613***	0.0704**	129.3***	0.0583**
	(1782)	(0.0281)	(36.90)	(0.0228)
Ethnic:				
Hispanic	2806	−0.0130	−123.8	−0.0876
	(5138)	(0.0920)	(103.5)	(0.0709)
Regional characteristics	Yes	Yes	Yes	Yes
House characteristics	Yes	Yes	Yes	Yes
Socio-economic variables	Yes	Yes	Yes	Yes
Device/equipment variables	Yes	Yes	Yes	Yes
No. of households	976	976	976	976
R-squared	0.639	0.673	0.656	0.679

We use all non-Black households as the baseline, and include all controls in the main model.
Robust standard errors adjusted by matching weights are in parentheses.
*p < 0.1, **p < 0.05, ***p < 0.01.

Our findings also contribute to the health literature by offering unique evidence on racial differences in space heating needs. Several empirical studies suggested that Black individuals exhibited a higher sensitivity to low temperatures than White individuals. For instance, Maley et al. (2014) conducted laboratory experiments with 30 participants and found that Black males exhibited greater sensitivity to cold compared to their Caucasian counterparts, while Asian males showed intermediate responses. Also, from a retrospective analysis of the British Army, Burgess and Macfarlane (2009) reported that Black males were 30.3 times more likely to experience cold injuries compared to Whites under comparable training, clothing, and nutrition conditions. Unlike these studies, which relied on a small sample or a retrospective analysis, our study provides large-scale evidence of

Table B.7
Racial differences in energy demand by end-use type among the matched sample based on propensity scores (N = 976).

Dependent variables:	Usage (kBTU/year)	ln(Usage)	Expenditure (US$/year)	ln(Expenditure)
End-use type:				
Space heating	2180*	0.225*	49.90**	0.156*
	(1323)	(0.128)	(21.67)	(0.0805)
Water heating	1717***	0.0938***	11.64	0.0229
	(441.7)	(0.0361)	(11.26)	(0.0336)
Air conditioning	474.3	−0.219	15.34	−0.0911
	(367.6)	(0.174)	(13.22)	(0.110)
Others	2242***	0.0177	52.44**	0.0113
	(723.6)	(0.0301)	(22.02)	(0.0279)

We use all non-Black households as the baseline, and include all controls in the main model.
Robust standard errors adjusted by matching weights are in parentheses.
*p < 0.1, **p < 0.05, ***p < 0.01.

Table B.8
Health outcomes among the matched sample based on propensity scores (N = 976).

Coefficient: Black	Medical attention needed because home was too cold	Medical attention needed because home was too hot
Sample:		
All	0.0174** (0.00795)	0.00428 (0.00562)
Low-income	0.0252** (0.0108)	0.00535 (0.00842)
High-income	−0.0103 (0.0135)	Not estimated

The coefficient for the high-income households among the matched sample was not estimated because none of them experienced medical attention due to too high indoor temperature.
Each entry represents the Black indicator's coefficient in the corresponding regression.
We use all non-Black households as the baseline, and include all controls in the main model.
All models are adjusted by sampling weights. Robust standard errors are in parentheses.
*p < 0.1, **p < 0.05, ***p < 0.01.

Table B.9
Racial disparities by climate based on the alternative criteria.

Coefficient: Black	Usage (kBTU/year)	Space heating (kBTU/year)	Air conditioning (kBTU/year)	Observations
Climate region:				
Hot-Humid	5395** (2495)	2061** (969.1)	129.2 (783.4)	1036
Dry/Mixed/Marine	4999** (1978)	3784*** (1467)	−391.6 (283.4)	2642
Cold	9495** (4121)	7244** (3240)	108.2 (239.1)	2008
Coefficient: Black	Expenditure (US$/year)	Space heating (US$/year)	Air conditioning (US$/year)	Observations
Climate region:				
Hot-Humid	82.82 (57.42)	34.72* (18.98)	6.264 (28.11)	1036
Dry/Mixed/Marine	106.2** (46.31)	68.28** (27.37)	−10.64 (10.48)	2642
Cold	120.5* (72.13)	91.10* (48.30)	5.287 (8.552)	2008

Climate regions are classified based on the Building America climate zone.
We use all non-Black households as the baseline, and include all controls in the main model.
Robust standard errors adjusted by sample weights are in parentheses. *p < 0.1, **p < 0.05, ***p < 0.01.

Table B.10
Racial differences by end-use type estimated by the inverse hyperbolic sine transformation.

Coefficient: Black	arcsinh(Usage)	arcsinh(Expenditure)
End-use type:		
Space heating	0.329*** (0.0954)	0.241*** (0.0638)
Water heating	0.0845*** (0.0295)	0.0511* (0.0268)
Air conditioning	−0.375** (0.163)	−0.210** (0.106)
Others	0.0188 (0.0275)	0.0105 (0.0252)

We use all non-Black households as the baseline, and include all controls in the main model.
Robust standard errors adjusted by matching weights are in parentheses.
*p < 0.1, **p < 0.05, ***p < 0.01.

Table B.11
Health outcomes estimated by alternative functional forms.

Coefficient: Black	Medical attention needed because home was too cold	Medical attention needed because home was too hot
Model:	Logistic regression	
Sample:		
All	1.182** (0.480)	−0.735 (0.886)
Low-income	1.342*** (0.495)	−0.516 (0.896)
High-income	Not estimated	Not estimated
Model:	Probit regression	
Sample:		
All	0.567*** (0.173)	−0.362 (0.261)
Low-income	0.635*** (0.180)	−0.287 (0.274)
High-income	Not estimated	Not estimated

The coefficients for the high-income households were not estimated due to perfectly predicting the outcomes using covariates.
Each entry represents the Black indicator's coefficient in the corresponding regression.
We use all non-Black households as the baseline, and include all controls in the main model.
Robust standard errors adjusted by sample weights are in parentheses.
*p < 0.1, **p < 0.05, ***p < 0.01.

the potential association between racial differences in cold sensitivity, energy consumption, and health outcomes. Moreover, we elucidate that this disparity correlates with a significant financial gap in the energy market.

In this vein, we call for discussions and solutions on this overlooked aspect of energy injustice. Although recent studies have raised concerns about the recognition injustice (White and Sintov, 2020; Walker and Day, 2012; Carley and Konisky, 2020; Ribó-Pérez et al., 2021), our understanding of energy vulnerability regarding this aspect is still nascent and not integrated yet. For instance, White and Sintov (White and Sintov, 2020) analyzed residents in warm and hot regions and found that Black households were not particularly energy-vulnerable than other groups in terms of health and financial outcomes. Although these results are consistent with our subgroup analysis by climate region, they can be adequately interpreted only after generalization efforts (e.g., examining regional heterogeneity) are sufficiently performed. We believe that further research and corresponding efforts to integrate it will enable policymakers and utility firms to design and implement more inclusive policies.

Notwithstanding our extensive efforts to absorb confounding factors, our approach could not completely rule out alternative explanations. Specifically, unobserved regional and house characteristics, such as solar orientation, might have biased the estimates. Furthermore, the current dataset does not provide information on expenditures on other items, such as food, housing rent, and transportation. Hence, we suggest future research to take account of households' various expenses to examine their economic decisions more thoroughly.

Moreover, our results for health outcomes should be interpreted cautiously due to the following limitations, which provide avenues for future research. Firstly, our understanding of health outcomes is restricted to self-reported needs for health services due to low indoor temperature. Future studies should integrate more specific health information into energy consumption data to further reveal the relationships among energy demand, indoor temperature, and health outcomes. Secondly, despite the observed association between space heating needs and self-reported health outcomes, it is still worth exploring why Black households consume more energy for space heating than other races. For instance, we do not show why health outcomes related to low indoor temperatures differ across racial groups, whereas those related to high temperatures do not. Lastly, although our analysis suggests a linkage between medical problems due to low indoor temperature and limited ability to pay for space heating, we could not directly discern the relationship between medical attention and space heating consumption due to the reverse causality problem. We suggest future research to investigate this relationship more closely.

CRediT authorship contribution statement

Sosung Baik: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Jeffrey F. Hines:** Conceptualization, Writing – original draft, Writing – review & editing. **Jaeung Sim:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Variable description

See Tables A.1–A.3.

Appendix B. Robustness checks

See Tables B.1–B.11.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eneco.2023.107098.

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